

DeepSequence: Hierarchical Attention Forecaster

DS + Softsign + Experts + Context-Aware Mixer

1. INPUTS

Heterogeneous feature groups



2. EXPERTS

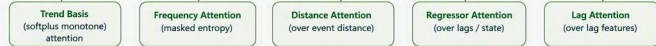
Specialized expert networks

(monotone: trend, holiday, regressor; seasonal: not monotone)



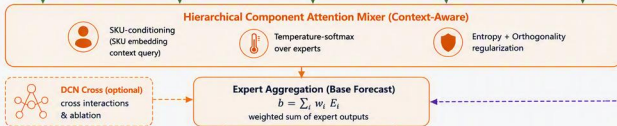
3. INTRA-EXPERT ATTENTION

Capture structure within each expert



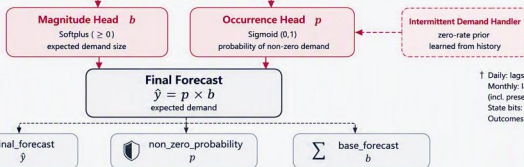
4. HIERARCHICAL MIXER

Context-aware mixing across experts



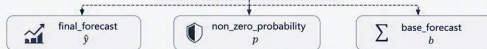
5. FORECAST HEADS

Decompose forecast into magnitude and occurrence



6. OUTPUTS

Final probabilistic forecast



Legend

- Inputs
- Experts & Intra-Expert
- Mixer & Aggregation
- Forecast Heads
- Outputs
- SKU Conditioning (conditioning path)
- Optional / Ablation Path

Notation

- E_i output of expert i
- w_i attention weight for expert i
- b magnitude (base forecast)
- p non-zero probability
- $\hat{y}$ final forecast

Notes

- Fourier uses fixed frequencies by default (learnable ω optional).
- Seasonal (Fourier) expert is not constrained to be monotonic.

† Daily: lags 1,2,7
Monthly: lags 1,2,12
(incl. present / min/max)
State bits: lag0 & 4
Outcomes: count & stable.

DeepSequence decomposes demand into occurrence (p) and magnitude (b), combines specialized experts (trend, seasonal/Fourier, holiday, regressor, lags) with hierarchical attention mixing, and forecasts intermittent and non-intermittent SKUs.